THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 2: **Power Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The value of $\frac{1}{2^{-3}}$ is: 1

(a) 8 **(b)** 6 **(c)** -8 **(d)** $\frac{1}{8}$

2. The value of $(-3)^4$ is: 1

(a) 81 **(b)** 85 **(c)** -81 **(d)** 12

3. The standard form of 60000000 is: 1

(a) 6×10^6 **(b)** 6×10^7 **(c)** 6×10^8 **(d)** 60×10^7

4. Which of the following equals 4.07×10^{-3}? 1

(a) 0.0407 **(b)** 4070 **(c)** 0.000407 **(d)** 0.00407

5. $\frac{1}{64}$ expressed as a power of 2 is: 1

(a) 2^{-6} **(b)** 2^{-5} **(c)** 2^6 **(d)** 2^{-7}

6. In 7^4, the base is: 1

(a) 4 **(b)** 28 **(c)** 2401 **(d)** 7

7. How many zeros are there in one lakh written in figures? 1

(a) 7 **(b)** 6 **(c)** 4 **(d)** 5

8. The value of $\frac{1}{3^{-2}}$ is: 1

(a) $\frac{1}{9}$ **(b)** 9 **(c)** 6 **(d)** -9

9. The value of $(-2)^5$ is: 1

(a) 10 **(b)** 37 **(c)** -32 **(d)** 32

10. The standard form of 384400 is: 1

(a) 3.844×10^5 **(b)** 3.844×10^4 **(c)** 3.844×10^6 **(d)** 38.44×10^5

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Express $(2^3)^4 \times 2^{-5}$ as a single power of 2. 2

12. Write $\left(\frac{2}{3}\right)^{-3}$ in simplest form. 2

13. The Earth is about 149600000 km from the Sun. Express it in standard form. 2

14. Express $(3^2)^5 \times 3^{-4}$ as a single power of 3. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Express each as a single power of 2: 3

(i) $2^5 \times 2^{-2}$

(ii) $(2^3)^{-2}$

(iii) $2^4 \div 2^{-3}$.

16. Find the value of $\left(\frac{2}{3}\right)^{-1} \times \left(\frac{2}{3}\right)^{-1}$. 3

17. Express the following in standard form: 3

(i) 5900000

(ii) 0.00074

(iii) 80000.

OR

Express each as a single power of 3: 3

(i) $3^4 \times 3^{-6}$

(ii) $(3^2)^{-3}$

(iii) $3^5 \div 3^{-2}$.

18. Find the value of $\left(\frac{3}{2}\right)^{-2} \times \left(\frac{3}{2}\right)^{-1}$. 3

OR

Express the following in standard form: 3

(i) 12300000

(ii) 0.0000058

(iii) 6500.

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Find x if 5

(i) $2^x \times 2^3 = 2^{11}$

(ii) $(2^x)^4 = 2^{20}$

(iii) $2^{x-2} = \frac{1}{2^3}$.

OR

Write 600000 and 2000 in standard form. Then find their product and quotient in standard form. 5

20. Find x if 5

(i) $3^x \times 3^2 = 3^9$

(ii) $(3^x)^5 = 3^{35}$

(iii) $3^{x-4} = \frac{1}{3^1}$.

OR

Write 80000000 and 400 in standard form. Then find their product and quotient in standard form. 5

END OF QUESTION PAPER Total: 40 Marks

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